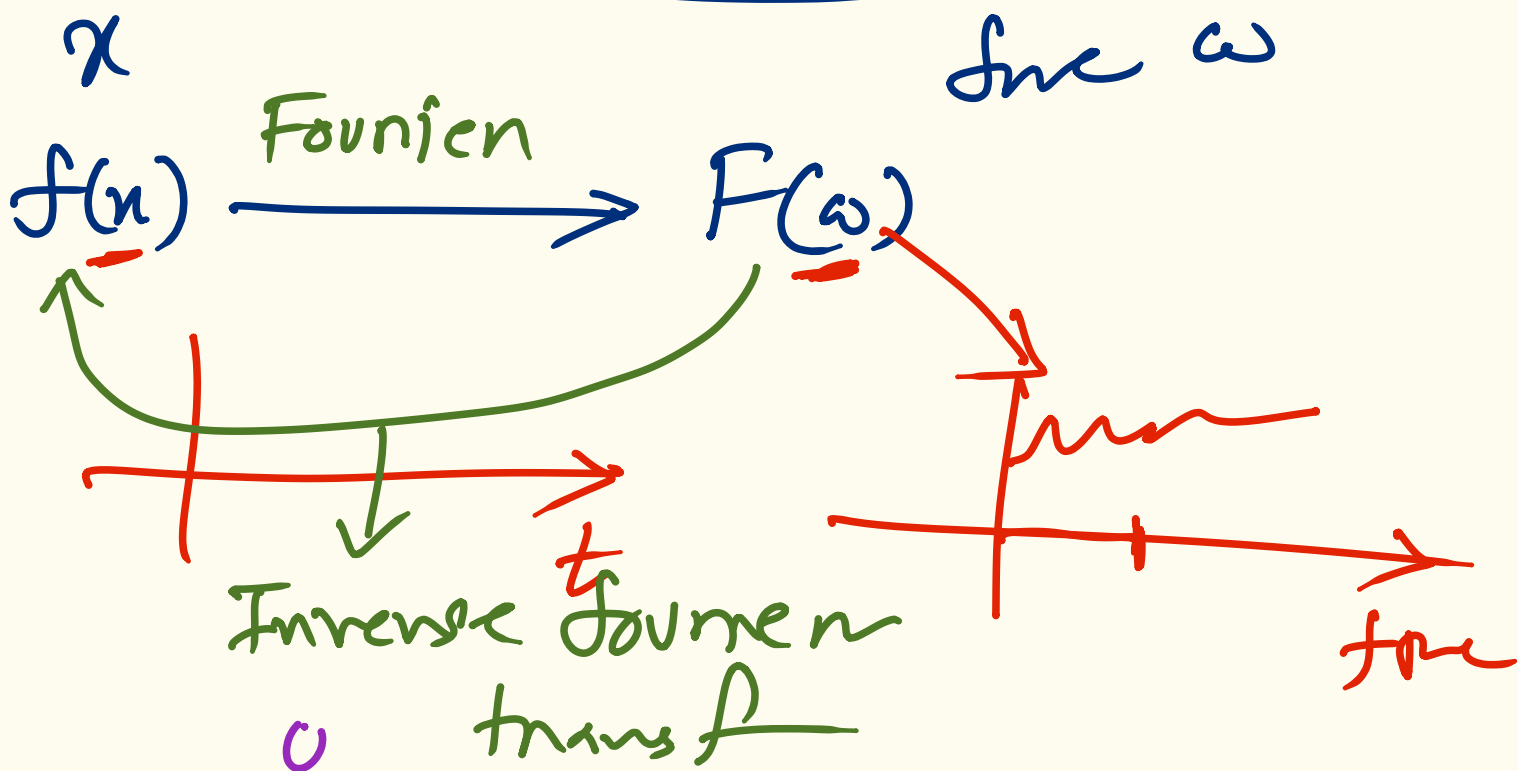




$$f(x) \sim \pi, \theta$$



$$\rightarrow \rho_1$$

$x \rightarrow \text{time}$   
 $\omega \rightarrow \text{frequency}$

## Fourier Transformation

## Fourier Transformation

Let  $f(x)$  be a continuous signal. The Fourier transformation of  $f(x)$  is denoted by  $F(\omega)$  defined by

$$\underline{e^{i\theta} = \cos \theta + i \sin \theta}$$

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \int_{-\infty}^{\infty} f(x) (\cos(-\omega x) + i \sin(-\omega x)) dx$$

$$= \int_{-\infty}^{\infty} f(x) \cos \omega x dx$$

$$- \int_{-\infty}^{\infty} f(x) \sin \omega x dx$$

$$\int_{-\infty}^{\infty} f(x) e^{i\omega x} dx$$

$$f(x) \xrightarrow[\text{transf}]{\text{Fourier}} F(\omega)$$

inverse  
Fourier transform

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

# Inverse Fourier Transformation

$F(\omega)$  be a frequency-domain signal. The Inverse Fourier transformation of  $F(\omega)$  is denoted by  $f(x)$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

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# Fourier Cosine Transformation

$f(x)$  continuous signal. The Fourier Cosine transformation of  $f(x)$  is denoted by  $F_c(\omega)$

$$F_c(\omega) = \int_0^{\infty} f(x) \cos(\omega x) dx$$

and inverse cosine transformation

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(\omega) \cos(\omega x) d\omega$$

# Fourier Sine Transformation.

$f(x)$  continuous signal. The Fourier sine transformation of  $f(x)$  is denoted by  $F_s(\omega)$

$$\checkmark F_s(\omega) = \int_0^{\infty} \underbrace{f(x)}_{\checkmark} \underbrace{\sin(\omega x)}_{\checkmark} dx$$

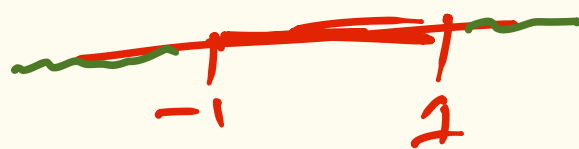
and inverse cosine transformation

$$\checkmark f(x) = \frac{2}{\pi} \int_0^{\infty} \underbrace{F_s(\omega)}_{\checkmark} \underbrace{\sin(\omega x)}_{\checkmark} d\omega$$

$$f(0) : \quad f(x) = \pi$$
$$f(0) = \pi$$

Find the Fourier transformation of the function  $f(x) = \begin{cases} \pi & |x| \leq 1 \\ 0 & |x| > 1 \end{cases}$   $\rightarrow -1 \leq x \leq 1$  hence  $(-\infty, -1), (1, \infty)$

evaluate  $\int_0^{\infty} \frac{\sin x}{x} dx$



F.T of  $f(x)$

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$

$$= \int_{-\infty}^{-1} f(x) e^{-i\omega x} dx + \int_{-1}^1 f(x) e^{-i\omega x} dx + \int_1^{\infty} f(x) e^{-i\omega x} dx$$

$$= 0 + \int_{-1}^1 \pi e^{-i\omega x} dx + 0$$

$$= \pi \int_{-1}^1 e^{-i\omega x} dx = \pi \left[ \frac{e^{-i\omega x}}{-i\omega} \right]_{-1}^1 = \frac{e^{kx}}{k}$$

$$= \frac{-\pi}{i\omega} [e^{i\omega \cdot 1} - e^{i\omega(-1)}]$$

$$= -\frac{\pi}{i\omega} [e^{-i\omega} - e^{i\omega}]$$

$$= \frac{\pi}{\omega} \left[ \frac{e^{i\omega} - e^{-i\omega}}{i} \right]$$

$$\therefore F(\omega) = \frac{\pi}{\omega} 2 \sin \omega$$

$$\therefore F(\omega) = \frac{2\pi}{\omega} \sin \omega$$

$$e^{i\theta} = \cos \theta + i \sin \theta \quad \text{--- (i)}$$

$$e^{-i\theta} = \cos \theta - i \sin \theta \quad \text{--- (ii)}$$

$$(i) - (ii)$$

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta$$

$$\Rightarrow 2 \sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{i}$$

$$\int_0^{\infty} \frac{\sin x}{x} dx = ?$$

Inverse Fourier transformation of  $F(\omega)$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \underline{F(\omega)} e^{i\omega x} d\omega$$

$$\Rightarrow f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\pi}{\omega} \sin \omega \cdot e^{i\omega x} d\omega$$

$$\Rightarrow f(x) = \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} e^{i\omega x} d\omega$$

$\Rightarrow$  Putting  $n=0$  in above eq

$$\Rightarrow f(0) = \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} e^0 d\omega$$

$$\Rightarrow \pi = \int_{-\infty}^{\infty} \frac{\sin \omega}{\omega} d\omega$$

$$\Rightarrow \pi = 2 \int_0^{\infty} \frac{\sin \omega}{\omega} d\omega$$

$$\Rightarrow \int_0^{\infty} \frac{\sin \omega}{\omega} d\omega = \frac{\pi}{2}$$

$$\boxed{\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}}$$

even  $f^n$

$$g(\omega) = \frac{\sin \omega}{\omega}$$

$$g(-\omega) = \frac{\sin(-\omega)}{-\omega}$$

$$= \frac{-\sin \omega}{-\omega}$$

$$= \frac{\sin \omega}{\omega} = g(\omega)$$

$$g(-\omega) = g(\omega)$$



Find Fourier Transformation of the function

$$f(x) = \begin{cases} 1, & |x| < a \rightarrow \underline{\underline{-a \leq x \leq a}} \\ 0, & |x| > a \end{cases}$$

hence evaluate  $\int_0^{\infty} \frac{\sin(ax) \cos(ax)}{x} dx$

Sol<sup>n</sup>: Fourier transformation of  $f(x)$

$$F(\omega) = \int_{-\infty}^{\infty} f(x) e^{-i\omega x} dx$$
$$= \int_{-a}^a 1 \cdot e^{-i\omega x} dx + 0$$

$$= \left[ \frac{e^{-i\omega x}}{-i\omega} \right]_{x=-a}^a$$

$$= \frac{e^{-i\omega a} - e^{i\omega a}}{-i\omega}$$

$$= \frac{e^{i\omega a} - e^{-i\omega a}}{i\omega}$$

$$= \frac{1}{\omega} \frac{e^{i\omega a} - e^{-i\omega a}}{i}$$

$$\therefore F(\omega) = \frac{1}{\omega} 2\sin(\omega a)$$

Taking inverse Fourier transformation

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega x} d\omega$$

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sin(\omega a)}{\omega} \cdot e^{i\omega x} d\omega$$

$$f(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(\omega a)}{\omega} [\cos \omega x + i \sin \omega x] d\omega$$

$$\Rightarrow \underline{f(x)} + \underline{i0} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(\omega a) \cos(\omega x)}{\omega} d\omega + \frac{i}{\pi} \int_{-\infty}^{\infty} \frac{\sin(\omega a) \sin(\omega x)}{\omega} d\omega$$

$$\checkmark \int_0^{\infty} \frac{\sin(\underline{ax}) \cos(\underline{ax})}{\underline{x}} \underline{dx}$$

The real parts of these should equal  
 so

$$f(x) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(\underline{\omega a}) \cos(\underline{\omega x})}{\underline{\omega}} \underline{d\omega}$$

by putting  $x=a$  in above equation

$$\underline{f(a)} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\sin(\omega a) \cos(\omega a)}{\omega} d\omega$$

$$\Rightarrow \underline{1} = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{\overset{\text{odd}}{\sin(\omega a)} \overset{\text{even}}{\cos(\omega a)}}{\omega_{\text{odd}}} d\omega$$

$$\Rightarrow 2 \int_0^{\infty} \frac{\sin(\omega a) \cos(\omega a)}{\omega} d\omega = \pi$$

$$\Rightarrow \int_0^{\infty} \frac{\sin(\omega a) \cos(\omega a)}{\omega} d\omega = \frac{\pi}{2}$$

$$\Rightarrow \int_0^{\infty} \frac{\sin(xa) \cos(xa)}{x} dx = \frac{\pi}{2}$$

$$\checkmark \checkmark F_s(\omega) = \int_0^{\infty} f(x) \sin(\omega x) dx$$

and inverse Cosine transformation

$$\checkmark \checkmark \checkmark f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin(\omega x) d\omega$$

Find the Fourier sine transformation of

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & x \geq 1 \end{cases}$$

hence evaluate

$$\int_0^{\infty} \frac{\sin^3 x}{x} dx$$

Sol<sup>n</sup>:

$$F_s(\omega) = \int_0^{\infty} \underline{f(x)} \sin(\omega x) dx$$
$$= \int_0^1 \sin(\omega x) dx + 0$$

$$= \left[ \frac{-\cos(\omega x)}{\omega} \right]_0^1$$

$$= -\frac{\cos \omega}{\omega} + \frac{\cos 0}{\omega}$$

$$F_s(\omega) = -\frac{\cos \omega}{\omega} + \frac{1}{\omega}$$

Taking Fourier inverse sine transformation,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(\omega) \sin(\omega x) d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \left( \frac{-\cos \omega}{\omega} + \frac{1}{\omega} \right) \sin(\omega x) d\omega$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{1 - \cos \omega}{\omega} \sin(\omega x) d\omega$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{2 \sin^2 \frac{\omega}{2}}{\omega} \sin(\omega x) d\omega$$

$$x = \frac{1}{2}$$

$$f\left(\frac{1}{2}\right) \Rightarrow \frac{2}{\pi} \int_0^{\infty} \frac{2 \sin^2 \frac{\omega}{2}}{\omega} \cdot \sin\left(\frac{\omega}{2}\right) d\omega$$

$$\Rightarrow 1 = \frac{4}{\pi} \int_0^{\infty} \frac{\sin^3 \frac{\omega}{2}}{\omega} d\omega$$

Let  $x = \frac{\omega}{2} \Rightarrow 2dx = d\omega$

|          |          |
|----------|----------|
| limit    |          |
| $\omega$ | $x$      |
| 0        | 0        |
| $\infty$ | $\infty$ |

$$\Rightarrow 1 = \frac{4}{\pi} \int_0^{\infty} \frac{\sin^3 x}{2x} \cdot 2dx$$

$$\Rightarrow \int_0^{\infty} \frac{\sin^3 x}{x} dx = \frac{\pi}{4}$$

